

Testing for statistically significant differences in performance between two quality estimator models

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When developing new models for speech quality estimation, it is necessary to compare the outputs from a given trained occurrence of a model with outputs from existing models or outputs from models with identical architecture but trained under different conditions or with different hyperparameters. In this case, we are specifically interested in determining when training a model with a lower number of votes per file causes statistically significant harm compared to a model trained with the least noisy data labels we have available. We use the exact the hypothesis test described in [1].

For a function y defined as

$$y = g(x_1, x_2, \dots, x_{2n}) \quad (1)$$

The combined uncertainty for y is given by [2] as

$$\begin{aligned} u_c^2(y) = & \sum_{i=1}^n \left(\frac{\partial g}{\partial x_i} \right)^2 u^2(x_i) \\ & + 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^n \frac{\partial g}{\partial x_i} \frac{\partial g}{\partial x_j} u(x_i) u(x_j) \text{corr}(x_i, x_j). \end{aligned} \quad (2)$$

Here we are interested in the difference between two mean Fisher-z transformed correlations, $\bar{z}_1, \bar{z}_2$ where

$$\bar{z}_j = \frac{1}{n_j} \sum_{i=1}^{n_j} z_{j,i}, \quad j = 1, 2, \quad (3)$$

so we define our measurement function as

$$y = g(z_{1,1}, \dots, z_{1,n_1}, z_{2,1}, \dots, z_{2,n_2}) = \bar{z}_1 - \bar{z}_2 = \frac{1}{n_1} \sum_{i=1}^{n_1} z_{1,i} - \frac{1}{n_2} \sum_{i=1}^{n_2} z_{2,i}. \quad (4)$$

Here we focus on the simple case with an equal number of tests for both experiments so $n = n_1 = n_2$, however the following can easily be done for the case $n_1 \neq n_2$. For convenience we define

$$z_i = \begin{cases} z_{1,i} & 1 \leq i \leq n \\ z_{2,i} & n+1 \leq i \leq 2n. \end{cases} \quad (5)$$

So we can write

$$g(z_1, z_2, \dots, z_{2n}) = \frac{1}{n} \sum_{i=1}^n z_i - z_{i+n}. \quad (6)$$

Due to the Fisher-z transformation we know that for a correlation value z_i , calculated over data with length n_f , that

$$u^2(z_i) \approx \frac{1}{n_f - 3}, \forall i, \quad (7)$$

and we define

$$\lambda_i = \frac{\partial g}{\partial z_i} = \begin{cases} \frac{1}{n} & 1 \leq i \leq n \\ -\frac{1}{n} & n+1 \leq i \leq 2n. \end{cases} \quad (8)$$

We can simplify (2) as

$$\begin{aligned} u_c^2(y) &= \sum_{i=1}^{2n} \frac{\lambda_i^2}{n_f - 3} + 2 \sum_{i=1}^{2n-1} \sum_{j=i+1}^{2n} \frac{\lambda_i \lambda_j}{n_f - 3} \text{corr}(z_i, z_j) \\ &= \frac{\sum_{i=1}^{2n} \lambda_i^2 + 2 \sum_{i=1}^{2n-1} \sum_{j=i+1}^{2n} \lambda_i \lambda_j \text{corr}(z_i, z_j)}{n_f - 3}. \end{aligned} \quad (9)$$

From [1] we know that the correlation between dependent Fisher's z transformed correlations z_i, z_j with $i \neq j$ is

$$\text{corr}(z_i, z_j) = r_x - \frac{r_i r_j}{2} + \frac{r_i r_j (r_i r_j - r_x)^2}{2(1 - r_i)^2 (1 - r_j)^2}, \quad (10)$$

where $z_i = \text{arctanh}(r_i)$. We can substitute the (10) into (9) in order to determine the combined uncertainty of difference of two Fisher's z transformed average correlation values.

References

- [1] X.-L. Meng, R. Rosenthal, and D. B. Rubin, "Comparing correlated correlation coefficients." *Psychological Bulletin*, vol. 111, no. 1, pp. 172–175, 1992, DOI: <https://doi.org/10.1037/0033-2909.111.1.172>.
- [2] BIPM, IEC, IFCC, ILAC, ISO, IUPAC, IUPAP, and OIML, "Evaluation of measurement data — Guide to the expression of uncertainty in measurement," Joint Committee for Guides in Metrology, JCGM 100:2008, DOI: <https://doi.org/10.59161/JCGM100-2008E>.